UT-free monotone polytopes without the star Ewald property

An explicit family with arbitrarily long lattice edges

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Abstract

For every integer $r \geq 2$, we construct a smooth reflexive polytope P_r of dimension $5r+1$ with $5r+4$ facets. Every vertex belongs to $-\mathbf{1} + r\mathbb{Z}^{5r+1}$, so every edge has lattice length divisible by r. In particular P_r has no unimodular triangular two-face. Nevertheless, an explicit face of dimension $2r-2$ fails the star Ewald condition: the active facet heights force a prospective symmetric lattice point to have one coordinate equal to -2 or 3. We also prove that P_r satisfies the strong Ewald condition and that its polar has full vertex minors of absolute values 2 and 3. The eleven-dimensional member therefore refutes the star Ewald assertion in Tu's Conjecture 2, and products give examples in every dimension at least eleven. The six-dimensional seed and the wedge operation have known antecedents in Paffenholz–McDuff and Haase–Melnikov. The additional result is an explicit extension retaining the obstruction while forcing arbitrary lattice edge divisibility. Historical priority for that extension is not certified here.

1 The exact target and the construction

A *monotone polytope* is a smooth reflexive lattice polytope, with smoothness understood in the Delzant sense: the primitive edge directions at each vertex form a lattice basis. A polytope is *UT-free* when none of its two-dimensional faces is a unimodular triangle. Tu's Conjecture 2 in [1], repeated in [2], asserts that UT-free monotone polytopes are duals of unimodular smooth Fano polytopes, and in particular satisfy the three Ewald conditions. We address the star assertion in its original form:

$$P \text{ monotone and UT-free} \quad \Longrightarrow \quad P \text{ is star Ewald.} \tag{1}$$

The 2024 text has an apparent extra word in the phrase describing the three conditions; the 2025 report states the three Ewald conditions without that ambiguity. Both sources explicitly define the star condition.

Fix an integer $r \geq 1$, and put

$$p = r+1, \qquad q = 2r, \qquad \ell = 2r-1, \qquad D = 5r+1.$$

Use coordinates

$$a \in \mathbb{R}^p, \quad b \in \mathbb{R}^q, \quad t \in \mathbb{R}^\ell, \quad s \in \mathbb{R}, \qquad A = \sum_i a_i, \quad B = \sum_j b_j, \quad T = \sum_\nu t_\nu.$$

Define

$$\boxed{P_r = \left\{ (a,b,t,s) : \begin{array}{l} a_i, b_j, t_\nu, s \geq -1, \\ T \leq 1, \quad A + s + 3T \leq 1, \quad B + s + 2T \leq 1 \end{array} \right\}.} \tag{2}$$

Theorem 1.1. *For every integer $r \geq 1$, the following statements hold.*

1. *P_r is D-dimensional, smooth and reflexive. It has $D + 3 = 5r + 4$ primitive irredundant facets and $4r^3 + 14r^2 + 7r$ vertices.*
2. *Every vertex lies in $-\mathbf{1} + r\mathbb{Z}^D$. Every edge has lattice length a positive multiple of r.*
3. *P_r is strong Ewald but is not star Ewald. An explicit failing face is*
$$F_r = \{a_i = b_j = -1,\ s = 4r - 1,\ T = 1 - r\} \cap P_r. \tag{3}$$
It is a simplex of dimension $2r - 2$, with edge length r when $r \geq 2$.
4. *The vertex configuration of the polar has full minors of absolute values 2 and 3, so it is not unimodular.*

Consequently P_r is UT-free for $r \geq 2$, and P_2 is an eleven-dimensional counterexample to (1).

All statements are proved below by explicit formulas valid for every r; computation is supplementary. This does not refute Ewald's original 1988 conjecture. Indeed we exhibit a lattice basis in $\mathcal{E}(P_r)$ and prove the stronger facet-by-facet basis property.

Attribution and scope. After a coordinate permutation, P_1 is the six-dimensional Paffenholz example discussed by McDuff [4, Section 2.1, Lemma 2.6]. Wedges and repeated facet splitting are established constructions [3, 4]. That seed has unimodular triangular faces. The additional conclusion here is preservation of its star obstruction together with $\mathrm{Vert}(P_r) \subset -\mathbf{1} + r\mathbb{Z}^D$. We do not claim a new wedge operation or the first non-star monotone polytope. Section 7.2 records the novelty boundary.

2 A facet-height criterion

For a monotone polytope $P = \{x \in \mathbb{R}^d : \langle n_i, x\rangle \leq 1,\ 1 \leq i \leq m\}$, with primitive integral outward normals, its *Ewald set* is

$$\mathcal{E}(P) = P \cap (-P) \cap \mathbb{Z}^d = \{x \in \mathbb{Z}^d : |\langle n_i, x\rangle| \leq 1 \text{ for all } i\}.$$

Weak Ewald means that $\mathcal{E}(P)$ contains a lattice basis. Strong Ewald means that $\mathcal{E}(P)$ on each facet contains a lattice basis.

For a face F, let $\mathrm{Star}(F)$ be the union of the facets containing F, let $\mathrm{star}(F)$ be the union of the ridges containing F, and put $\mathrm{Star}^*(F) = \mathrm{Star}(F) \setminus \mathrm{star}(F)$. The face is star Ewald if some $\lambda \in \mathcal{E}(P)$ satisfies

$$\lambda \in \mathrm{Star}^*(F), \qquad -\lambda \notin \mathrm{Star}(F).$$

The polytope is star Ewald when every nonempty proper face is. These are the definitions used in [1, 5].

Lemma 2.1. *Let $I(F)$ index the facets containing F. A point $\lambda \in \mathcal{E}(P)$ is a star witness for F if and only if*

$$\big(\langle n_i, \lambda\rangle\big)_{i \in I(F)}$$

has exactly one entry equal to 1, with all other entries zero.

Proof. The entries are integers in $\{-1, 0, 1\}$. The condition on $-\lambda$ excludes -1, and membership in $\mathrm{Star}(F)$ requires a 1. In a simple polytope, two distinct facets containing F intersect in a ridge containing F. Membership in $\mathrm{Star}^*(F)$ therefore allows exactly one 1. The converse is immediate. □

This criterion will exclude every possible witness, without a height cutoff or a search through a finite portion of the lattice.

3 A fixed pentagon describes every vertex

Set

$$a_i = r\alpha_i - 1, \quad b_j = r\beta_j - 1, \quad t_\nu = r\tau_\nu - 1, \quad s = r\sigma - 1.$$

This is a vertex description, not a claimed unimodular equivalence. It gives $P_r = rR_r - \mathbf{1}$, where

$$R_r = \left\{ \begin{array}{l} \alpha_i, \beta_j, \tau_\nu, \sigma \geq 0, \quad U := \sum_\nu \tau_\nu \leq 2, \\ \sum_i \alpha_i + \sigma + 3U \leq 7, \quad \sum_j \beta_j + \sigma + 2U \leq 6 \end{array} \right\}. \tag{4}$$

The projection to (σ, U) is the pentagon

$$K = \{(\sigma, U) : \sigma \geq 0,\ 0 \leq U \leq 2,\ \sigma + 3U \leq 7,\ \sigma + 2U \leq 6\},$$

whose vertices are

$$(0,0),\ (6,0),\ (4,1),\ (1,2),\ (0,2).$$

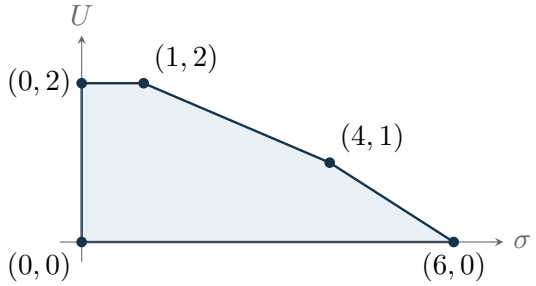


Figure 1: The pentagon is fixed. Increasing r changes the numbers of coordinates in its simplex fibers.

Over (σ, U), the fiber is the product of

$$\begin{aligned} &\alpha \geq 0, \quad \sum_i \alpha_i \leq L_a := 7 - \sigma - 3U, \\ &\beta \geq 0, \quad \sum_j \beta_j \leq L_b := 6 - \sigma - 2U, \\ &\tau \geq 0, \quad \sum_\nu \tau_\nu = U. \end{aligned}$$

Lemma 3.1. *The vertices of R_r are exactly the fiber vertices over the five vertices of K. In particular, every vertex of R_r is integral.*

Proof. Include the slack coordinates $L_a - \sum \alpha_i$ and $L_b - \sum \beta_j$. Barycentric proportions describe each fiber point. If its base point lies in a nontrivial segment of its minimal face in K, keep those proportions fixed and vary the base point. The fiber sizes are affine, so this produces an affine segment through the given point in R_r. A zero fiber size causes no exception: a nonnegative affine function vanishing at a relative interior point of a face vanishes on that face. Thus a vertex projects to a vertex of K, and it must be a fiber vertex. Conversely, any convex decomposition of a fiber vertex over a base vertex stays in that fiber and is trivial. A fiber vertex allocates its whole size to one coordinate, possibly the slack coordinate. The base vertices and fiber sizes are integral. □

The sum is $4r^3 + 14r^2 + 7r$. The same description proves

$$\mathrm{Vert}(P_r) \subset -\mathbf{1} + r\mathbb{Z}^D. \tag{5}$$

(σ, U)	L_a	L_b	U	Number of vertices
$(0,0)$	7	6	0	$(r+2)(2r+1)$
$(6,0)$	1	0	0	$r+2$
$(4,1)$	0	0	1	$2r-1$
$(1,2)$	0	1	2	$(2r+1)(2r-1)$
$(0,2)$	1	2	2	$(r+2)(2r+1)(2r-1)$

Table 1: Complete vertex enumeration for every $r \geq 1$.

4 Smoothness, reflexivity, and UT-freeness

The system (4) is bounded. Every inequality in (2) is strict at the origin, giving full dimension.

Lemma 4.1. *Exactly D facets meet at each vertex of P_r, and their primitive outward normals have determinant ± 1.*

Proof. Work in R_r, whose facet normals are the same. First eliminate the zero coordinates with their lower unit normals. If an α or β coordinate is positive, it is the only positive coordinate in its block; its upper inequality supplies a unit pivot in that coordinate. When $U > 0$, precisely one τ coordinate is positive.

The five cases in Table 1 leave the following residual matrices, respectively:

$$(), \qquad (1), \qquad \begin{pmatrix} 1 & 3 \\ 1 & 2 \end{pmatrix}, \qquad \begin{pmatrix} 1 & 3 \\ 0 & 1 \end{pmatrix}, \qquad (1).$$

The empty determinant is 1; all others are ± 1. In the middle case the remaining coordinates are σ and the positive τ coordinate, and the two upper inequalities supply the displayed matrix. At $(1,2)$, the first upper inequality and $U = 2$ remain after any β carrier is eliminated. The other cases have only triangular unit pivots. The list accounts for every active inequality and has exactly D independent rows in each case. □

Every lower-coordinate inequality is a facet: set that coordinate to -1 and all others to zero. For the A upper facet, use $a_1 = 1$ and all other coordinates zero; the analogous point works for the B upper facet. For $T = 1$, take all $t_\nu = 1/\ell$ and all a_i, b_j, s slightly greater than -1, making both other upper inequalities strict. All $D + 3$ inequalities are therefore irredundant; their normals are primitive.

The vertices are integral by Lemma 3.1. All primitive facet equations of P_r have constant 1, and the origin is interior. The polar is thus a convex hull of integral normals and is integral, proving reflexivity. Lemma 4.1 proves smoothness and hence monotonicity.

For distinct edge endpoints v, w, the lattice length is the greatest common divisor of the coordinates of $v - w$, with signs and zero entries ignored. Equation (5) makes it a positive multiple of r. For $r \geq 2$, every edge contains an interior lattice point. A unimodular triangle has primitive edges, so P_r is UT-free.

5 The star obstruction and polar determinants

The face F_r is obtained by setting every $a_i, b_j = -1$ and making the two upper inequalities involving A, B equalities. Solving gives $s = 4r - 1$, $T = 1 - r$. The remaining coordinates satisfy

$$t_\nu + 1 \geq 0, \qquad \sum_{\nu=1}^{2r-1} (t_\nu + 1) = r.$$

Thus F_r is a $(2r-2)$-simplex, of edge length r for $r \geq 2$. At its relative interior $t_\nu > -1$, $s > -1$, and $T < 1$. Its containing facets are exactly the a, b lower facets and the two upper facets involving A, B.

If $\lambda \in \mathcal{E}(P_r)$ were a star witness, Lemma 2.1 would make all their heights zero except for one height 1. Group the a lower facets with the A upper facet, and the b lower facets with the B upper facet. The two sums of heights, ε, δ, obey

$$(\varepsilon, \delta) \in \{(1,0), (0,1)\}.$$

Cancellation in the groups gives

$$s(\lambda) + 3T(\lambda) = \varepsilon, \qquad s(\lambda) + 2T(\lambda) = \delta.$$

Consequently

$$s(\lambda) = 3\delta - 2\varepsilon \in \{-2, 3\}. \tag{6}$$

But both $\lambda, -\lambda$ satisfy $s \geq -1$, so $|s(\lambda)| \leq 1$, a contradiction. This proves the star failure for every r.

For the polar, use $P_r^\circ = \{y : \langle y, x\rangle \leq 1 \text{ for all } x \in P_r\}$. Its vertices are the primitive facet normals. In the negative coordinate matrix $-I_D$, replace one t_ν row by the normal of $A + s + 3T \leq 1$. The determinant has absolute value 3. Replacing it by the normal of $B + s + 2T \leq 1$ gives absolute value 2. A unimodular coordinate change preserves these absolute determinants. The selected full subsets are not facets of the polar; smoothness requires unit determinants for actual facets, whereas the USFP definition requires them for every full subset.

Tu's negative-polar convention gives $-P_r^\circ$. The map $-I$ is unimodular, so the convention has no effect here.

6 Strong Ewald is preserved

A direct weak Ewald basis is

$$\{e_{a_i}\}_i \cup \{e_{b_j}\}_j \cup \{e_s\} \cup \{e_{t_\nu} - e_{a_1} - e_s\}_\nu. \tag{7}$$

Elementary column operations give determinant 1. Every vector has all facet heights in $\{-1, 0, 1\}$. For the last type, the three upper heights are $1, 1, 1$. Thus the construction explicitly satisfies Ewald's original condition.

Lemma 6.1 (Strong Ewald under facet splitting)**.** *Let $Q = \{x : n_i x \leq 1\}$ be monotone and strong Ewald, with the $n_i x \leq 1$ its primitive irredundant facet inequalities. Replace each inequality by*

$$z_{i,j} \geq -1 \ (1 \leq j < k_i), \qquad n_i x + \sum_{j<k_i} z_{i,j} \leq 1,$$

where the k_i are positive integers. If the resulting polytope L is monotone and these are its facets, then it is strong Ewald.

Proof. Fix a top facet in block i, and choose a lattice basis $x_1, \ldots, x_d$ in $\mathcal{E}(Q) \cap \{n_i x = 1\}$. Embed these vectors with new coordinates zero. For every new coordinate in block i, add its unit vector. For a new coordinate in a different block g, add $(x_1, \eta e_{g,j})$, where $\eta = \pm 1$ is chosen so that $n_g x_1 + \eta \in [-1, 1]$. Such a choice exists because $n_g x_1 \in \{-1, 0, 1\}$. All these vectors are Ewald points on the chosen top facet. Subtracting the first embedded vector when appropriate leaves a block triangular basis of determinant ± 1.

The integral linear involution $z_{i,j} \mapsto -n_i x - \sum_h z_{i,h}$, with other coordinates fixed, preserves L, has determinant -1, and exchanges the top facet with the corresponding lower facet. It transfers the basis to every lower facet. □

Lemma 6.2. *The six-dimensional member P_1 is strong Ewald.*

Proof. Use seed coordinates $(a_1, a_2, b_1, b_2, s, t_1) = (x_1, x_2, x_3, x_4, x_5, x_6)$. The upper normals are

$$e_6, \qquad c = (1, 1, 0, 0, 1, 3), \qquad d = (0, 0, 1, 1, 1, 2).$$

Put $w = e_6 - e_1 - e_5$. For $x_i = -1$, $1 \le i \le 5$, use the six vectors

$$-e_i, \quad \{-e_i + e_j : 1 \le j \le 5,\ j \ne i\}, \quad u_i,$$

where $u_i = w$ for $i = 1, 5$, and $u_i = w - e_i$ otherwise. Every normal height is in $\{-1, 0, 1\}$. The first five vectors form a basis in the first five coordinates, and the last has sixth coordinate 1, so the determinant is ± 1. For $x_6 = 1$, use $v, v + e_1, \ldots, v + e_5$, where $v = (-1, -1, -1, -1, -1, 1)$. These are Ewald points of determinant ± 1. Their negatives handle $x_6 = -1$. Finally the separate involutions

$$x_1 \mapsto -x_1 - x_2 - x_5 - 3x_6, \qquad x_3 \mapsto -x_3 - x_4 - x_5 - 2x_6$$

with other coordinates fixed preserve P_1, have determinant -1, and exchange the corresponding lower and upper facets. This treats all nine facets. □

Split the lower seed facets for x_2, x_3, x_4, x_6 into blocks of sizes $r, r, r, 2r - 1$. For a split lower facet $-x_i \le 1$, the coordinates

$$z_{i,1}, \ldots, z_{i,k_i - 1}, \quad x_i - \sum_j z_{i,j}$$

all have lower bound -1, sum to x_i, and are obtained by a unimodular transformation. Regrouping them makes the old coordinates

$$x_1 = a_1, \quad x_2 = \sum_{i=2}^{r+1} a_i, \quad x_3 = \sum_{j=1}^{r} b_j, \quad x_4 = \sum_{j=r+1}^{2r} b_j, \quad x_5 = s, \quad x_6 = T.$$

The resulting inequalities are exactly (2). Monotonicity was independently proved above. Lemmas 6.1 and 6.2 now prove strong Ewald for every P_r, completing Theorem 1.1.

7 Consequences, exact checks, and prior results

Corollary 7.1. *For every dimension $N \ge 11$ there is a UT-free monotone polytope that is strong Ewald and not star Ewald. No dimension-independent lower bound on lattice edge length forces a monotone polytope to be star Ewald.*

Proof. Use $P_2 \times [-1, 1]^{N-11}$. Products preserve smoothness and reflexivity, and every edge has length at least 2. A star witness for $F_2 \times [-1, 1]^{N-11}$ would project to one for F_2.

Strong Ewald is preserved under products: for a facet $H \times Q$, embed an Ewald basis on H, and append one fixed member of that basis paired with each member of an Ewald basis of Q. The resulting basis is contained in that facet. The other facet type is symmetric. The final assertion follows by taking r arbitrarily large in Theorem 1.1. □

We do not claim that eleven is the minimum possible dimension. The last assertion concerns dimension-independent bounds only.

7.1 Exact verification

For $r = 2$, all $\binom{14}{11} = 364$ candidate normal bases were tested with exact rational arithmetic. The resulting list agrees with the analytic description. Separate implementations, including one in split coordinates, gave the same result.

Quantity	Exact value
Dimension	11
Primitive irredundant facets	14
Vertices	102
Edges	561
Edges of lengths $2, 4, 12, 14$	216, 255, 60, 30
Vertices of the bad face	3
Absolute active vertex determinants	1
Exhibited polar minors	2 and 3

These checks supplement the all-r proof. Appendix A supplies a compact standard-library verifier. The manuscript is not a proof-assistant formalization and has not received external peer review.

7.2 The novelty boundary

The known seed is not UT-free: its face $x_3 = x_4 = x_5 = -1$, $x_6 = 1$ is the unit triangle with free-coordinate vertices $(-1, -1), (0, -1), (-1, 0)$. Products retain such a face. The present construction changes the dimension and facet multiplicities while retaining monotone normalization, and forces all vertices into one residue class modulo r.

A separate, simpler overlap affects the dual-USFP inclusion alone. The known simplex–segment bundle

$$|x| \le 1, \quad y_i \ge -1 \ (1 \le i \le 3), \quad 2x + y_1 + y_2 + y_3 \le 1$$

is UT-free by [5, Proposition 5.14]. Its polar has a determinant-two minor, as in the family of [6, Example 7.1]. These facts already imply a counterexample to the inclusion, but do not give a non-star example: the SSB family satisfies star Ewald [5, Proposition 5.16]. The additional claim examined here is specifically the UT-free, non-star family with arbitrary edge divisibility.

The primary-source audit inspected the conjecture, its 2025 repetition, the wedge antecedents, CPS, and the three available six-dimensional examples on Paffenholz's page [7]. Each of those three has a unimodular triangular face. The linked seven-dimensional archive was not retrievable. Equivalent descriptions involving long edges, Fano index, and probe failure were also considered. No inspected source supplied the combined all-r conclusion. This is a bounded search, not proof of priority.

Criterion	Established scope
G1	The star assertion occurs in Tu's prior Conjecture 2.
G2	The original UT-free, monotone, and star definitions are unchanged.
G3	The cited sources propose the conjecture; historical absence of an earlier resolution is not certified.
G4	The counterexample and the full r-family have unconditional analytic proofs.
G5	Vertex congruence and a surviving height obstruction yield an actual new-result candidate using known wedges. A historically new fundamental method is not claimed.
G6	Known overlapping constructions are attributed. First-discovery status remains unverified.

A A compact exact verifier

The following Python 3 code needs no third-party packages. It verifies the $r = 2$ vertices, simplicity, smoothness, parity, edge lengths, bad face, Ewald basis, and nonunit polar minors. The analytic proof additionally establishes facet irredundancy and every statement for all r.

```
from fractions import Fraction as F
from itertools import combinations
from collections import Counter
from math import gcd
from functools import reduce

r = 2
p, q, h = r+1, 2*r, 2*r-1
D = p+q+h+1
ia = list(range(p))
ib = list(range(p, p+q))
it = list(range(p+q, p+q+h))
S = D-1

def unit(j):
    v = [0]*D
    v[j] = 1
    return v

def dot(a, b):
    return sum(x*y for x, y in zip(a, b))

def solve(M, rhs):
    n = len(M)
    B = [[F(x) for x in row] + [F(rhs[i])]
         for i, row in enumerate(M)]
    det = F(1)
    for j in range(n):
        k = next((k for k in range(j, n) if B[k][j]), None)
        if k is None:
            return None, F(0)
        if k != j:
            B[k], B[j] = B[j], B[k]
            det = -det
        z = B[j][j]
        det *= z
        B[j] = [x/z for x in B[j]]
        for k in range(n):
            if k != j:
                z = B[k][j]
                B[k] = [x-z*y for x, y in zip(B[k], B[j])]
    return tuple(B[i][-1] for i in range(n)), det

N = [[-x for x in unit(j)] for j in range(D)]
nt = [int(j in it) for j in range(D)]
na = [int(j in ia)+int(j == S)+3*int(j in it)
      for j in range(D)]
nb = [int(j in ib)+int(j == S)+2*int(j in it)
      for j in range(D)]
N += [nt, na, nb]

vertices = {}
for J in combinations(range(D+3), D):
    v, det = solve([N[j] for j in J], [1]*D)
    if v is None or any(dot(n, v) > 1 for n in N):
        continue
    assert all(x.denominator == 1 for x in v)
    active = tuple(i for i, n in enumerate(N) if dot(n, v) == 1)
    assert len(active) == D and abs(det) == 1
    vertices[tuple(map(int, v))] = frozenset(active)

assert len(vertices) == 102
assert all(all((x+1) % r == 0 for x in v) for v in vertices)
assert set().union(*vertices.values()) == set(range(D+3))
```

```
edges = Counter()
for v, w in combinations(vertices, 2):
    if len(vertices[v] & vertices[w]) == D-1:
        length = reduce(gcd, (abs(x-y) for x, y in zip(v, w)))
        edges[length] += 1
assert edges == Counter({2:216, 4:255, 12:60, 14:30})

bad = [v for v in vertices
       if all(v[j] == -1 for j in ia+ib)
       and v[S] == 4*r-1 and sum(v[j] for j in it) == 1-r]
assert len(bad) == 3
active_bad = frozenset(ia+ib+[D+1,D+2])
assert frozenset.intersection(*(vertices[v] for v in bad)) == active_bad

ga = [sum(N[i][j] for i in ia+[D+1]) for j in range(D)]
gb = [sum(N[i][j] for i in ib+[D+2]) for j in range(D)]
assert ga == [int(j == S)+3*int(j in it) for j in range(D)]
assert gb == [int(j == S)+2*int(j in it) for j in range(D)]

basis = [unit(j) for j in ia+ib+[S]]
for j in it:
    w = unit(j)
    w[ia[0]] = -1
    w[S] = -1
    basis.append(w)
_, det = solve(list(map(list, zip(*basis))), [0]*D)
assert abs(det) == 1
assert all(abs(dot(n, w)) <= 1 for n in N for w in basis)

for upper, expected in [(na,3), (nb,2)]:
    M = [row[:] for row in N[:D]]
    M[it[0]] = upper
    _, det = solve(M, [0]*D)
    assert abs(det) == expected

print("ALL EXACT CHECKS PASSED")
print("vertices =", len(vertices), "edges =", sum(edges.values()))
print("edge length histogram =", dict(sorted(edges.items())))
```

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